

# William Sander

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## Professional Summary

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Technical support and software development professional with a B.S. in Computer Science and experience resolving hardware, software, data, and configuration issues across **3,000+ mobile devices**. Built production automation in Python, C++, and SQL for the U.S. Census Bureau, reducing manual work and improving operational reliability. Experienced with Linux, IoT hardware, REST APIs, machine learning systems, and AI-assisted development workflows; known for methodical troubleshooting, ownership, and clear cross-team communication.

## Education

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University of Vermont

**B.S. Computer Science**

*Selected Coursework:* Machine Learning, Advanced Programming (C++), Algorithms, Data Structures, Operating Systems (C), Computability and Complexity, Distributed Computing

Burlington, VT

Aug 2023 - May 2026

Community College of Vermont

**Associate's Degree, STEM Studies**

Winooski, VT

Aug 2021 - May 2023

## Relevant Projects

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Python | PyTorch | Data Analysis | Model Diagnostics | Flask

**Speech Emotion Recognition System** — [GitHub](#), UVM, Advanced Machine Learning

- Built an ML investigation pipeline across **8 multilingual datasets** and **~25.8k samples**, using evaluation metrics to identify performance patterns and compare model behavior
- Developed diagnostic and interpretability tooling with attention maps and Grad-CAM, making incorrect or unexpected system outputs easier to investigate
- Deployed Flask live inference at **10–20ms latency** and achieved **81.29% accuracy** and **83.63% F1** on the 4-class evaluation

C++ | OpenGL | Python | NumPy | Systems Debugging

**C++ Canvas Raster Graphics Editor** — [Demo](#) · [GitHub](#), UVM, Advanced Programming

- Built and debugged a real-time graphics system with **7 tools**, multi-layer buffers, and diff-based undo/redo; maintained **60 FPS** while keeping memory use under **140 MB**
- Investigated performance and integration issues across C++, OpenGL, the Python C API, and NumPy while implementing image filters and lazy CPU-to-GPU texture updates

IoT | Python | Embedded Linux | REST APIs | Hardware Integration

**E-Ink iCloud Calendar Display** — [GitHub](#)

- Built and troubleshooted an end-to-end IoT system on Raspberry Pi and a **7.5"** e-paper display, integrating hardware, embedded Linux, iCloud data, and public REST APIs
- Developed modular Python software to validate and merge ICS/JSON inputs, handle inconsistent upstream data, and render a reliable unattended display

## Work Experience

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U.S. Census Bureau

**IT Clerk (Technical Support and Automation)**

Burlington, VT

Feb 2020–Nov 2020

- Investigated and resolved hardware, software, account, and configuration issues for **hundreds of field staff** and **3,000+ mobile devices**, taking ownership through resolution during time-sensitive operations
- Analyzed device records and operational data to identify configuration drift, custody discrepancies, and recurring failure patterns; built validation tooling that improved data accuracy and reliability
- Developed production automation in **Python, C++, and SQL** to replace repetitive reconciliation and device-custody workflows, increasing support capacity during the COVID-extended census timeline
- Collaborated with regional IT and operations teams to diagnose cross-system issues, communicate technical findings, and deploy fixes across distributed infrastructure
- Documented repeatable solutions and procedures so staff could resolve common issues consistently and escalate complex cases with useful diagnostic context

## Skills

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**Support / Operations:** Technical troubleshooting, root-cause analysis, issue investigation, workflow automation, documentation, cross-team escalation

**Scripting / Data:** Python, SQL, C++, C, R, Java

**Systems / Tools:** Linux, Git, Docker, REST APIs, Flask, VSCode

**AI / Hardware:** AI agents, PyTorch, model evaluation, Raspberry Pi, IoT, e-paper displays, embedded peripherals